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ORIGINAL INVESTIGATION

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CT attenuation values and mineral distribution can be used to differentiate dogs with and without gallbladder mucoceles

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Abstract

Gallbladder mucoceles are potentially fatal in dogs. Multiphase CT angiography was performed to evaluate the canine gallbladder in three conditions: no sludge, sludge occupying ≥25% of the lumen, and mucoceles. Twenty dogs with normal hepatobiliary bloodwork and no-to-minimal gallbladder sludge, 13 dogs with normal bloodwork and ≥25% sludge in the gallbladder lumen, and 18 dogs with histologically confirmed gallbladder mucoceles were enrolled in a prospective, observational diagnostic accuracy study. Three regions of interest (ROI) were stratified in the dorsal-ventral orientation and a single ROI was measured within the hepatic parenchyma. Mean attenuation and presence of mineral were recorded. Average Hounsfield units (HU) were recorded for precontrast, arterial, portovenous, and late venous phases. The overall median HU value for mucoceles was significantly higher than gallbladders without sludge and with sludge; precontrast median overall attenuation was 49.3, 35.8, and 39.7 HU, respectively ($P < .000004$). Mineral was seen in four (20%) dogs with no sludge, seven (56%) dogs with sludge, and nine (50%) dogs with mucoceles. Mineral in the dogs with mucoceles was located within the central aspect of the gallbladder lumen in 67% of mucoceles; this mineral distribution was not seen in any dog without a mucocele. Computed tomography can differentiate a subset of gallbladder mucoceles from dogs with and without gallbladder sludge, especially in the precontrast series. An HU value of 48.6 is 52% sensitive and 96% specific for a gallbladder mucocele. A hyperattenuating gallbladder on precontrast CT images and centrally distributed mineral can be a gallbladder mucocele.

KEYWORDS
canine, cholecystolithiasis, gallbladder mineral, sludge

1 | INTRODUCTION

Gallbladder mucoceles are a potentially fatal disease in dogs characterized by hyperplasia of the mucosal epithelium with resulting increased mucin secretion.[1–3] Morbidities related to gallbladder mucoceles include rupture of the gallbladder, infection, or extrahepatic biliary duct obstruction with potential to result in systemic inflammatory response syndrome and bile peritonitis. Gallbladder sludge is a more common finding with less serious pathologic implications. Biliary sludge is described in ultrasound as gravity-dependent echogenic luminal contents without distal acoustic shadowing; the significance is unknown.[1–5] Recent studies have questioned the role of biliary sludge in the onset of pathology as no direct causation has been found.[1–3,6–8] The clinical implications comparing gallbladder mucoceles and biliary sludge are vastly different. Gallbladder mucoceles are life-threatening, commonly requiring immediate cholecystectomy, whereas sludge may require only medical intervention or monitoring.[5,9–14] Ultrasound has been the conventional tool for diagnosis of gallbladder mucoceles and differentiation from biliary sludge; however, with the growing utilization of CT, the appearance of a potentially fatal gallbladder disease is not known.

Ultrasound is the current standard for diagnosis of gallbladder mucoceles.[11] The appearance of the diseased gallbladder can vary but is typically characterized by radiating or stellate hyperechoic striations within immoveable, hypoechoic gallbladder contents (classically referred to as a "kiwi" pattern).[1,8,15] Pathogenesis of gallbladder mucoceles is unclear and multiple possibilities have been postulated.[7,11] Cocker spaniels, miniature schnauzers, and Shetland sheepdogs appear particularly prone to gallbladder mucocele development and a hereditary component is considered in these breeds.[9,15–18]

Abbreviations: AUC, area under the curve; ROI, Region of interest.

Recent research has also shown the migratory capacity of extruded gallbladder mucoceles in the abdomen as identified on abdominal ultrasound.[19] Although mucoceles are readily detected using ultrasonography, relative to CT, it can result in longer scan times, greater patient discomfort from physical compression of the ultrasound probe, and incomplete evaluation of the abdomen in large patients.[20] Thus, CT could allow faster and more complete abdominal evaluation for these patients, but scientific investigation of canine biliary contents in veterinary medicine using CT is minimal.[21] Computed tomography is becoming a more accessible tool and, in some instances, a first choice diagnostic modality for larger dogs.[20,22] The ability of CT to distinguish between canine gallbladder sludge and mucoceles is unknown.

The aims of this study were to describe the imaging characteristics of the gallbladder in three conditions (no sludge, sludge occupying ≥25% of the lumen, and mucoceles) using CT to evaluate the utility of attenuation values, or Hounsfield units (HU), for characterizing gallbladder contents. It was hypothesized that quantifying HU using multiphase CT angiography would not result in significantly different HU between gallbladder mucoceles, gallbladder sludge, and dogs without gallbladder sludge due to the subtle attenuation changes between soft tissue on CT.

2 | MATERIALS AND METHODS

Privately owned dogs were prospectively recruited for an observational diagnostic accuracy study from clients of The Ohio State University Veterinary Medical Center between 2016 and 2018. Medical histories and physical examinations were performed on all dogs and serum chemistry and ultrasound of the gallbladder were used to place patients into one of three groups. Subject inclusion for the study was determined by an American College of Veterinary Radiology-certified veterinary radiologist (E.T.H.) after evaluation of a screening ultrasound and systemic bloodwork that consisted of hepatobiliary markers. The no-sludge group had hepatobiliary bloodwork (ALP, AST, GGT, and total bilirubin) within the normal reference ranges with < 25% gallbladder sludge on screening ultrasound, the sludge group had hepatobiliary bloodwork within the normal reference ranges with ≥25% sludge in the gallbladder lumen on screening ultrasound, and the mucocele group had histologically confirmed gallbladder mucoceles following either cholecystectomy or necropsy with supportive hepatobiliary bloodwork. Dogs were excluded from the no-sludge group and the ≥25% sludge groups if ALP, AST, GGT, and/or total bilirubin were above the reference range. Dogs were excluded from the mucocele group if histopathology did not determine a gallbladder mucocele. Informed client consent was obtained for all dogs and the study was approved by the Institutional Animal Care and Use Committee of The Ohio State University (2016A00000097) as well as the Clinical Research and Advising Committee of The Ohio State University Veterinary Medical Center.

Sedated or anesthetized helical CT angiography studies (64-detector GE Revolution EVO, GE Healthcare, Waukesha, WI) were performed in each dog from midway through the heart to the coxofemoral joints. All dogs were positioned in sternal recumbency and no breath holds were used as not all dogs were intubated. Sedation/anesthesia protocols differed and were determined by the clinician with direct responsibility of the patient according to their preference; anesthetic monitoring (pulse oximetry, electrocardiography) was performed in all patients while imaging. No attempt was made to standardize heart rate or blood pressure. Scan parameters included a variable automatic filament current with a maximum milliamperage of 500 mA, a fixed maximum voltage of 120 kVp, pitch 0.984, rotation time 0.6 s, and slice thickness 1.25 mm. Although reformatted images in dorsal and sagittal planes were generated for review of the study, these images were not included in data analysis and were not critical to the study design. Iodinated contrast medium (Omnipaque 240™ Iohexol injection, GE Healthcare, Princeton, NJ) was administered via a cephalic catheter to all dogs at a dose of 2 mg/kg using a power injector (Mark V Plus, Bayer, Whippany, NJ) set to 3 mL/s injection rate with a maximum pressure of 300 psi. The same time interval was used between contrast phases in each dog; after the precontrast series, the arterial phase was initiated using bolus-tracking software with a region of interest (ROI) placed in the aorta at the level of the celiac artery; the portovenous phase occurred 20s after the start of the arterial phase; and late venous phase imaging occurred 60 s after the start of the arterial phase. The established timing protocol was used for ease of repeatability. Though no contrast enhancement was expected within the gallbladder lumen itself, postcontrast data were used chiefly for comparison to adjacent liver parenchyma. The presence/absence of gallbladder rupture was not recorded during surgery or necropsy in all dogs and, therefore, CT data regarding gallbladder rupture or wall thickness were not included.

Images were reviewed by a third-year veterinary radiology resident (J.A.F.) using a commercial DICOM image viewer (eFilm Workstation 3.2, Merge Healthcare, Chicago, IL). A standard algorithm in a soft tissue window display (W 400, L 40) was used to assess the same CT slice in the precontrast, arterial, portovenous, and late venous phases; the slice that was selected for data analysis had the greatest subjective gallbladder cross-sectional area. The gallbladder was divided into a dorsal, middle, and ventral third based off of dorsoventral height on the selected slice; three ROIs of identical area (0.3 cm^2) were placed in the gallbladder, one ROI in each third (Figure 1). If hyperattenuating material was seen within the gallbladder lumen, it was included in the ROI measurements as long as its attenuation did not exceed 100 HU, which was considered mineral for this study. An additional ROI was placed within hepatic parenchyma adjacent to the gallbladder on the same slice, avoiding large vasculature. Mean attenuation within each ROI was recorded as was the presence of any mineral (> 100 HU) within the gallbladder; if present, the shape and position of the mineral was recorded. All measurements were acquired by a third-year radiology resident (J.A.F.) with guidance from an American College of Veterinary Radiology-certified veterinary radiologist (E.T.H.). The observers were not blinded to the grouping of each dog since objective values were recorded with a predetermined technique for the observational study. Ratios of the gallbladder luminal content attenuation to hepatic parenchyma attenuation were then calculated for each ROI within

FIGURE 1 Portovenous phase postcontrast transverse CT image (window width: 400, window length: 40) of the liver and gallbladder. Three regions of interest (black rings) were placed over the gallbladder lumen in the dorsal, middle, and ventral thirds with an additional region of interest (white ring) over hepatic parenchyma on the same image

each contrast phase. Finally, overall median HUs across each phase were calculated by including all ROI measurements within each group.

Statistical analysis was performed by an author with biostatistical training as part of an advanced degree (E.T.H.) using commercially available software (MedCalc Software bvba, Ostend, Belgium; https://www.medcalc.org; 2016). The data were analyzed for normality using the Shapiro-Wilk test. Due to mixed normal and non-normal distribution between groups, nonparametric data analysis was performed. A Kruskal-Wallis ANOVA with post hoc Jonckheere Terpstra trend test was used to compare gross HU and ratios between each of the three gallbladder groups under different variables. A repeated measures ANOVA was used to evaluate differences within each group for the four vascular phases (precontrast, arterial, portovenous, and late venous), as well as when comparing the dorsal, middle, and ventral strata within each group. A Fisher's exact test was performed to evaluate the central location of the mineral compared to the dependent location in dogs with a gallbladder mucocele and dogs without a mucocele (consisting of the sludge group and no sludge group). A P-value $\leq .05$ was considered significant.

3 | RESULTS

Fifty-one dogs were included in the study; 20 dogs were included in the no-sludge group, 13 dogs in the sludge group, and 18 dogs in the mucocele group. The 33 dogs without mucoceles had a median age of 7 years (range 1-11 years) with a median weight of 28.2 kg (range 4.9-44.6 kg), 16 were castrated male, 14 were spayed female, and three were intact male. Breeds included mixed breed (9), Miniature Dachshund (2), Golden Retriever (2), Pit Bull (2), Labrador Retriever (2), and one each of Shilo Shepherd, Standard Poodle, German Shepherd, Gordon Setter, Rottweiler, Alaskan Malamute, American Bulldog, Boxer, Shar Pei, Australian Cattle Dog, Beagle, Miniature Schnauzer, Miniature Poodle, Smooth Fox Terrier, Shih Tzu, and Pug. Median (range) biochemistry values were ALT 35 (15-62), ALP 32 (7-114), GGT 3 (0-7), and total bilirubin 0.08 (0.01-0.22); two dogs did not have AST reported and that value was not included in calculations.

The 18 dogs with gallbladder mucoceles had a median age of 11 years (range 3-14 years) with a median weight of 9.4 kg (range 5-42.2 kg), six were castrated male, 10 were spayed female, one was intact male, and one was intact female. Breeds included mixed breed (4), Cocker Spaniel (2), Shih Tzu (2), Pomeranian (2), and one each of Rough Collie, Shetland Sheepdog, Australian Cattle Dog, Beagle, West Highland White Terrier, Miniature Schnauzer, Miniature Dachshund, and Pug. Median (range) biochemistry values were ALT 959 (29-3552), ALP 2902 (112-8840), GGT 32.5 (3-177), and total bilirubin 1.51 (0.05-13.26).

Median overall HU between each group in each phase are reported in Table 1. When comparing overall median HU, mucocele dogs were significantly different from the biliary sludge and no-sludge groups across all precontrast, arterial, portovenous, and late venous phases ($P < .00001$) with the most significant difference present on the precontrast phase ($P < 0.000004$). On the portovenous and late venous phases, the median attenuation value for sludge dogs significantly differed from dogs without sludge ($P < .00001$); there was no significant difference between these groups on the remaining phases.

When comparing the hepatic ROIs in dogs without mucoceles, all phases significantly differed from one another regardless of group or phase. In dogs with mucoceles, the portovenous and late venous phases were significantly different from the precontrast and arterial phases but not different from each other; all other phases were significantly different.

When comparing between the three strata of the gallbladder within each group, there was no significant difference between any layer for mucocele dogs ($P \geqq 0.11$). For dogs with sludge, the ventral stratum differed from the dorsal on precontrast, arterial, and portovenous phases but not on the late venous phase ($P < 0.03$); there was no difference between strata on the late venous phase ($P = .09$). In dogs without sludge, the dorsal layer was significantly different from the middle and ventral layers across all phases except for venous where it only significantly differed from the ventral layer ($P < 0.01$).

Sensitivity, specificity, and receiver operating characteristic curves were generated for each phase using HU to compare attenuation values of the gallbladder within the group of mucoceles to the two groups without mucoceles. The analysis was performed as the one group of mucoceles compared to the two groups of gallbladders with and without sludge to address the clinical question of detecting a mucocele. For precontrast, the area under the curve (AUC) was greatest at 0.731 with a Youden index of 0.4781 at a cutoff value of 48.6 HU yielding a sensitivity and specificity of 51.8% and 95.9%, respectively.

TABLE 1 Median overall Hounsfield units (HU) (range) for each group in each phase of contrast

	Precontrast	Arterial	Portovenous	Late venous
No sludge	35.8 (11.2–48.6)	36.5 (11.2–46.5)	36.2 (11.7–50.5)	39.1 (9.4–52.5)
Sludge	39.7 (13.5–83.0)	38.4 (10.2–91.2)	38.7 (14.1–84.5)	41.1 (19.9–88.3)
Mucocele	49.3 (22.8–87.3)*	54.6 (18.0–92.2)*	48.7 (22.1–92.1)*	53.1 (13.1–88.0)*

*indicates significantly higher HU compared to the no sludge and sludge groups.

TABLE 2 Threshold Hounsfield unit (HU) values for receiver operating characteristic curve analysis comparing raw attenuation of gallbladder mucoceles to non-mucoceles displaying the high specificity and low sensitivity of CT

	Area under curve	*P*-value	Youden Index	Cut-off (HU)	Sensitivity (%)	Specificity (%)
Pre-Contrast	0.731	<.001	0.4781	>48.6	51.85	95.96
Arterial	0.730	<.001	0.5101	>44.5	61.11	89.90
Portovenous	0.710	<.001	0.4949	>46.5	55.56	93.94
Late Venous	0.691	<.001	0.4663	>51.6	53.70	92.93

TABLE 3 Threshold gallbladder Hounsfield unit (HU) to hepatic parenchyma HU ratio for receiver operating characteristic curve analysis comparing CT diagnosis of gallbladder mucoceles versus non-mucoceles

	Area under curve	*P*-value	Youden index	Cut-off ratio	Sensitivity (%)	Specificity (%)
Pre-Contrast	0.736	<.0001	0.5236	>0.87	57.41	94.95
Arterial	0.615	.024	0.3283	>0.80	38.89	93.94
Portal	0.579	.128	0.2761	>0.58	40.74	86.87
Venous	0.710	<.001	0.4680	>0.49	51.85	94.95

The HU cutoff value provides a threshold attenuation to aid in the screening of gallbladder mucoceles. An HU value above 48.6 would be a true positive for a mucocele in 51.8% of the cases, whereas if the HU is below 48.6 then no mucocele is present in 95.9%. For the arterial phase, the AUC was 0.730 with a Youden index of 0.5101 at a cutoff of 44.5 HU yielded a sensitivity and specificity of 61.1% and 89.9%, respectively. The AUC for the portovenous and late venous phases were lower (Table 2). An receiver operating characteristic analysis was also performed on the gallbladder HU to hepatic parenchyma HU ratios; the results are reported in Table 3.

Mineral was present in four (20%) of the dogs with no sludge, seven (54%) of the dogs with sludge, and nine (50%) of the dogs with mucoceles. The mineral attenuating material was seen in the gravity-dependent aspect of the lumen in all 13 dogs (100%) without mucoceles. The mineral material was seen in the gravity dependent aspect of mucocele gallbladders in three of nine dogs with the remaining six dogs (67%) having amorphous mineral positioned within the central, gravity independent aspect of the lumen; the six dogs with centrally located mineral attenuating material were all dogs with mucoceles. The central location of mineral was significantly associated with gallbladder mucoceles ($P = .0011$) based on the Fisher exact test.

4 | DISCUSSION

Findings from the current study indicated that CT angiography can be used to identify a subset of dogs with gallbladder mucoceles. Gallbladder mucoceles had significantly higher HU during all vascular phases compared to the dogs with and without sludge. Also, mucoceles were significantly higher regardless of the stratified level at which the ROI was placed. Although a majority of mucoceles are significantly hyperattenuating relative to normal gallbladders, there is overlap of the attenuations with multiple mucoceles having attenuation values similar to gallbladders without mucoceles (Figure 2). The greatest specificity and sensitivity for identifying gallbladder mucoceles using CT angiography is at a cutoff of 48.6 HU; there is a high specificity of 96% with a low sensitivity of 52%.

The presence and, more importantly, the distribution of mineral within the gallbladder lumen can aid in the identification of mucoceles. All dogs without mucoceles that had mineral in the gallbladder had it in the gravity-dependent aspect of the lumen and it was considered to be representative of cholecystolithiasis. Two-thirds of dogs with mucoceles had mineral material within the central aspect of the gallbladder lumen independent of gravity (Figure 3). All dogs with centrally positioned mineral had mucoceles. The central mineral occasionally assumed a radiating or stellate shape reminiscent of the appearance of a mucocele on ultrasound. Despite this trend, three mucoceles still had mineral in the gravity-dependent aspect of the gallbladder lumen that was similar in appearance to the non-mucocele dogs.

The receiver operating characteristic curves display the use of HU for diagnosing gallbladder mucoceles. The AUC was greatest for both raw HU and gallbladder to hepatic parenchyma ratio in the precontrast phase (0.731 and 0.736, respectively). The shape of the

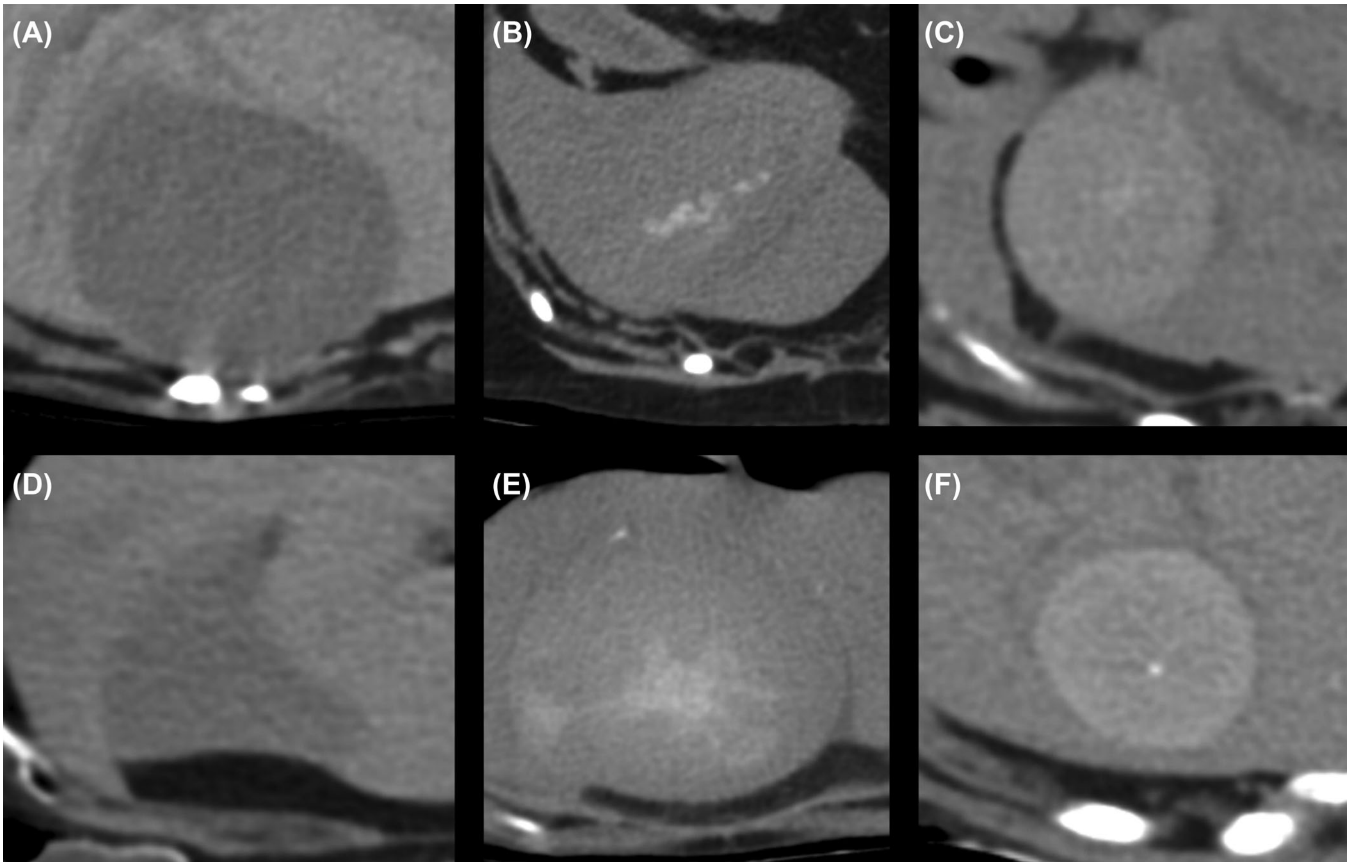

FIGURE 2 Precontrast transverse CT images (window width:400, window length:40) of the gallbladder in sternally positioned dogs with mucoceles. The variation in mucocele appearance is highlighted with examples of mucoceles that are hypoattenuating (A, D), isoattenuating (B, E), and hyperattenuating (C, F) to hepatic parenchyma. Also note the centrally positioned mineral in dogs (B), (C), (E), and (F)

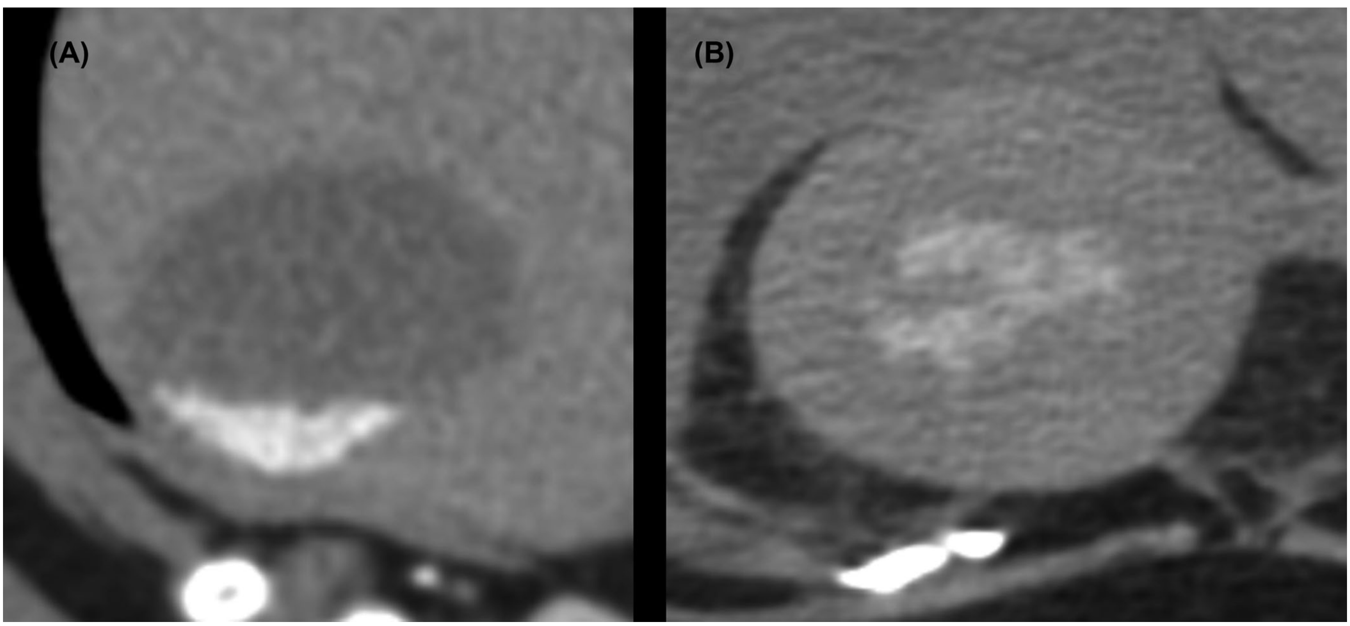

FIGURE 3 Precontrast transverse CT images (window width: 400, window length: 40) of the gallbladder in sternally positioned dogs with sludge (A) and a mucocele (B). Note the ventral, gravity-dependent mineral in the dog with sludge contrasted with the centrally positioned gallbladder mineral in the dog with a mucocele

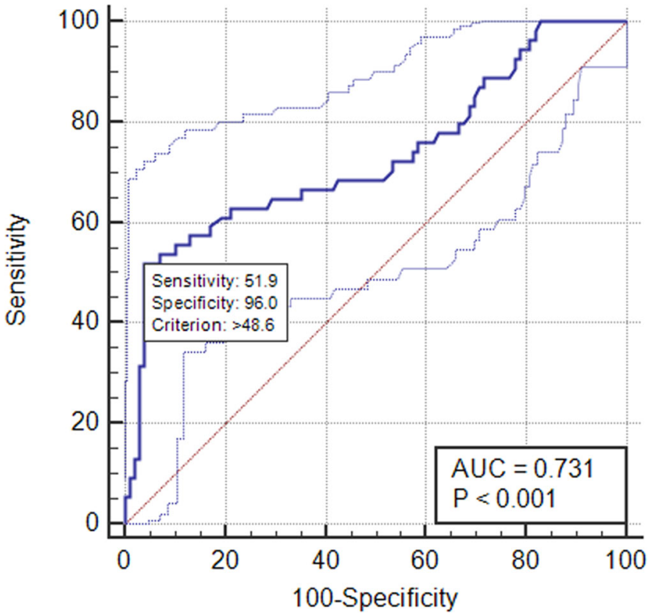

FIGURE 4 The receiver operating characteristic of the precontrast phase Hounsfield Unit attenuation of mucoceles versus non-mucoceles displaying the low sensitivity and high specificity of CT for mucocele detection [Color figure can be viewed at wileyonlinelibrary.com]

receiver operating characteristic curve displays that CT angiography is highly specific but has a low sensitivity for the diagnosis of gallbladder mucoceles. An HU value below the thresholds in Table 2 is unlikely to be associated with a gallbladder mucocele; however, HU values greater than the threshold increase the concern for a gallbladder mucocele for approximately 50% of cases (Figure 4). For the non-contrast series, a ratio of the gallbladder HU to the hepatic parenchyma HU > 0.87 had a 57.4% sensitivity and a 95.0% specificity for diagnosing a gallbladder mucocele. For some dogs with gallbladder mucoceles on ultrasound, the changes present are relatively subtle and do not translate to a significant increase in luminal attenuation values on CT thus allowing for the presence of false negatives. With the consistent hypoattenuating appearance of the gallbladder in the dogs without mucoceles, the appearance of a hyperattenuating gallbladder on CT should result in infrequent false positive mucocele identification.

The gallbladder luminal contents stratify in a gravity-dependent fashion in dogs without mucoceles. Stratification of gallbladder luminal contents is not observed in dogs with mucoceles. The stratification of denser materials freely moving within a bladder lumen is also observed in the urinary bladder.[23] The content in gallbladders for dogs without mucoceles was relatively hyperattenuating ventrally compared to dorsally. No significant stratification was statistically provable in dogs with mucoceles. The mild variation in the stratification seen between the groups without mucoceles is likely due to the small study population. Despite the variation of the appearance of gallbladder mucoceles on CT, subjectively the authors noted a trend within the mucocele group of the gallbladder lumen being diffusely hyper- to isoattenuating to hepatic parenchyma on precontrast and arterial images, respectively, and the lumen becoming hypoattenuating to hepatic parenchyma by the venous phases. The gallbladder in the non-mucocele groups was consistently hypoattenuating to the hepatic parenchyma.

There are limitations within the study to acknowledge. The authors acknowledge that there is a spectrum of treatment options and differing opinions; however, as histopathology was a requirement for inclusion, there was an inherent bias within the mucocele group toward more severe mucoceles. Any mucocele that received medical management alone (typically more mild clinical cases) was excluded from the study. Histopathology of the gallbladder was not performed in either of the no-sludge or sludge groups and there is potential that these dogs could have had underlying biliary disease that was not reflected in systemic bloodwork. The lack of breath-holds during CT also allowed for respiratory motion artifact to be present between contrast phases resulting in slight variation between the slices evaluated. Authors attempted to recognize the variation between vascular phases and synchronize the anatomy when applicable. The timing of the vascular phases was not tailored to the size of the dog and the sedation protocol was determined by the primary clinician. The standardization for an imaging protocol triggered by an HU threshold at a consistent location and variation in sedation could have resulted in variation of wash-in and wash-out hemodynamics for the liver. The authors feel ratios of the gallbladder ROI to the hepatic ROI would be less reliable than raw attenuation values for this reason. Future studies can include the prospective analysis of dogs with acute abdomen using the criteria determined in this study. Additional blinded observers of differing experience levels can review the current cases to evaluate for interobserver and intraobserver repeatability for the use of CT in cases of gallbladder mucoceles. Future enrollment of cases will emphasize the presence of gallbladder rupture to improve the diagnostic use of CT for gallbladder mucoceles in the presence of rupture. Histopathology and chemical composition analysis of the gallbladder mucocele contents can help to determine why a variation of Hounsfield units is identified.

This study puts into perspective the use of CT angiography in the evaluation of canine gallbladders and its use for the diagnosis of gallbladder mucoceles. The central, amorphous distribution of mineral within the gallbladder lumen supports hyperplasia of the mucosal epithelium restricting a peripheral distribution of potential lumen space. The combination of a hyperattenuating gallbladder with centrally distributed mineral during a precontrast CT accompanied by increased hepatobiliary markers on blood work is consistent with a gallbladder mucocele. Although authors recommend ultrasonography as a standard imaging modality for dogs with a high suspicion of mucocele, this study supports the use of CT angiography as an adjunct imaging modality for the diagnosis of gallbladder mucoceles.

LIST OF AUTHOR CONTRIBUTIONS

Category 1

(a) Concept and Design: Hostnik

(b) Acquisition of Data: Fuerst

(c) Analysis and Interpretation of Data: Hostnik, Fuerst

Category 2

(a) Drafting the Article: Fuerst

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Category 3

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